

MP1994V2 Proof Ledger

Mortensen–Pissarides (1994) Lean replication

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Purpose. The milestone tracker records project state. This cumulative ledger records exactly what Lean has proved, the hypotheses used, and the economic meaning of each theorem.

1 M1: Surplus Bellman equation

Paper reference	Section 3, journal page 401, equation (8)
Adequacy status	COMPLETE – GREEN
Lean source	MP1994V2/SteadyState/Surplus.lean

1.1 Mathematical statement

For the actual equilibrium surplus

$$S(\varepsilon) = J(\varepsilon) + W(\varepsilon) - U,$$

Equation (8) is a conditional representation theorem: for every object satisfying **ValueEquilibrium** **P**, Lean proves that the surplus derived from its value functions satisfies, for every idiosyncratic state ε ,

$$(r + \lambda)S(\varepsilon) = p + \sigma\varepsilon - b + \lambda \int \max\{S(x), 0\} dF(x) - \beta\theta q(\theta)S(\varepsilon_u). \quad (8)$$

Here **P.shock** is the probability law of the idiosyncratic shock; its induced CDF is F , and integration with respect to dF is represented in Lean by integration against **P.shock**. Also, $\theta q(\theta)$ is **P.workerMeetingRate** **E.theta**. M1 does not prove existence of a **ValueEquilibrium**.

1.2 Economic interpretation

- $p + \sigma\varepsilon$: current match output.
- b : unemployment opportunity cost.
- $\lambda \int \max\{S(x), 0\} dF(x)$: option value of future idiosyncratic draws, allowing rejection of nonpositive continuation surplus.
- $\beta\theta q(\theta)S(\varepsilon_u)$: the employed worker’s foregone search gain at the upper-support job.

1.3 Exact Lean declarations

Role	Declaration
Continuation decomposition	MP1994V2.continuationTerm_eq_integral_activeSurplus_sub
Firm surplus share	MP1994V2.ValueEquilibrium.firm_share
Algebraic surplus flow	MP1994V2.ValueEquilibrium.surplus_flow_equation
Minimal probability theorem	MP1994V2.ValueEquilibrium.surplus_bellman_of_probability
Paper-level wrapper	MP1994V2.ValueEquilibrium.surplus_bellman

1.4 Assumptions

Used:

- equation (3), through the definition of actual surplus;
- equation (4), only at the upper state;
- equations (5) and (6), added to cancel the wage;
- equation (7), subtracted to account for unemployment value;
- probability normalization of `P.shock`;
- `ValueEquilibrium.active_surplus_integrable`.

The minimal theorem assumes only `[IsProbabilityMeasure P.shock]`. The wrapper takes `D : ShockAssumptions P`, installs `D.isProbability` locally, and uses no other field of `D`.

Not used: parameter signs; `CoreEconomicAssumptions`; upper-support restrictions or maximality; atomlessness; mean zero; unit variance; `ShockNormalizationAssumptions`; matching positivity or monotonicity; `MatchingAssumptions`; equations (1) and (2).

1.5 Readable proof

Write the shared continuation term as

$$C(\varepsilon) = \int [\max\{S(x), 0\} - S(\varepsilon)] dF(x).$$

Adding equations (5) and (6) cancels the wage and sums the Nash weights:

$$r[J(\varepsilon) + W(\varepsilon)] = p + \sigma\varepsilon + \lambda C(\varepsilon).$$

Subtracting equation (7), and using equation (3), gives

$$rS(\varepsilon) = p + \sigma\varepsilon - b + \lambda C(\varepsilon) - \theta q(\theta)[W(\varepsilon_u) - U].$$

Equation (4) at ε_u gives

$$W(\varepsilon_u) - U = \beta S(\varepsilon_u).$$

Probability normalization of `P.shock` and active-surplus integrability give

$$C(\varepsilon) = \int \max\{S(x), 0\} dF(x) - S(\varepsilon).$$

Substitution and moving $-\lambda S(\varepsilon)$ to the left proves (8).

1.6 Adequacy and next dependency

The actual `ValueEquilibrium` surplus is used. Equation (8) is derived, not added as an assumption or field. No cutoff, reservation rule, or affine surplus result is used. It is conditional on an existing `ValueEquilibrium` and does not establish that such an equilibrium exists.

Human-review conclusion. Equation (8) matches the paper and is derived non-circularly from equations (3)–(7). The actual equilibrium surplus is used, with no cutoff, affinity result, or reduced equilibrium. The theorem is explicitly conditional on an existing `ValueEquilibrium`.

M2 uses (8) as the reviewed dependency for the affine-surplus and reservation results recorded next.

Review space:

2 M2: Affine surplus and endogenous reservation cutoff

Paper reference	Section 3; journal p. 401, reservation-property paragraph; journal p. 402, equation (12)
Adequacy status	COMPLETE – GREEN
Lean sources	MP1994V2/SteadyState/AffineSurplus.lean; MP1994V2/SteadyState/Cutoff.lean

2.1 Mathematical statements

For every existing `ValueEquilibrium` `P`, subtracting equation (8) at two states gives

$$(r + \lambda) [S(\varepsilon_2) - S(\varepsilon_1)] = \sigma(\varepsilon_2 - \varepsilon_1).$$

Under $r > 0$, $\lambda \geq 0$, and $\sigma > 0$,

$$S(\varepsilon_2) - S(\varepsilon_1) = \frac{\sigma}{r + \lambda}(\varepsilon_2 - \varepsilon_1),$$

so S is strictly increasing. Define

$$\varepsilon_d = \varepsilon_u - \frac{(r + \lambda)S(\varepsilon_u)}{\sigma}.$$

Lean proves a unique zero on all of $\mathbb{R}$:

$$S(\varepsilon_d) = 0, \quad S(\varepsilon) = 0 \iff \varepsilon = \varepsilon_d.$$

The reservation rules are

$$\begin{aligned} S(\varepsilon) < 0 &\iff \varepsilon < \varepsilon_d, & 0 \leq S(\varepsilon) &\iff \varepsilon_d \leq \varepsilon, \\ J(\varepsilon) < 0 &\iff \varepsilon < \varepsilon_d, & 0 \leq J(\varepsilon) &\iff \varepsilon_d \leq \varepsilon. \end{aligned}$$

Paper equation (12) and its zero-anchored form are

$$S(\varepsilon) - S(\varepsilon_d) = \frac{\sigma(\varepsilon - \varepsilon_d)}{r + \lambda}, \quad S(\varepsilon) = \frac{\sigma}{r + \lambda}(\varepsilon - \varepsilon_d).$$

The M3 bridge is

$$\max\{S(\varepsilon), 0\} = \frac{\sigma}{r + \lambda} \max\{\varepsilon - \varepsilon_d, 0\}.$$

2.2 Proof strategy and economic separation

The expectation and upper-job search term in equation (8) are common across current states, so they cancel when the two equation instances are subtracted. The resulting positive affine slope constructs the unique mathematical zero.

Economic admissibility is proved separately. Equation (2) and $r > 0$ give $V = 0$; equation (1) then gives

$$q(\theta)J(\varepsilon_u) = c.$$

Positive vacancy cost and meeting imply $J(\varepsilon_u) > 0$. Since $1 - \beta > 0$, the upper-state surplus is positive, and therefore $\varepsilon_d < \varepsilon_u$. Recruiting cost thus places the reservation state strictly below the productivity of a newly formed upper-support job.

2.3 Exact Lean declarations

Role	Declaration
Scaled two-point identity	ValueEquilibrium.surplus_difference_scaled_of_probability
Divided affine identity	ValueEquilibrium.surplus_difference
Strict monotonicity	ValueEquilibrium.surplus_strictMono
Derived cutoff	ValueEquilibrium.reservationCutoff
Cutoff zero	ValueEquilibrium.surplus_reservationCutoff_eq_zero
Unique zero	ValueEquilibrium.existsUnique_surplus_zero
Vacancy value	ValueEquilibrium.vacancy_value_eq_zero
Free-entry product	ValueEquilibrium.vacancy_meeting_mul_upper_firm_value_eq_cost
Upper firm value	ValueEquilibrium.upper_firm_value_pos
Upper surplus	ValueEquilibrium.upper_surplus_pos
Cutoff below upper point	ValueEquilibrium.reservationCutoff_lt_epsUpper
Equation (12)	ValueEquilibrium.equation12
Surplus sign rule	ValueEquilibrium.surplus_neg_iff_lt_reservationCutoff
Firm sign rule	ValueEquilibrium.firm_value_neg_iff_lt_reservationCutoff
Active-surplus bridge	ValueEquilibrium.activeSurplus_eq_slope_mul_positivePart
Capstone	ValueEquilibrium.milestone2_capstone

Capstone zero identification. The capstone explicitly includes

$$\forall \varepsilon, \quad S(\varepsilon) = 0 \iff \varepsilon = \varepsilon_d,$$

rather than merely asserting an unidentified unique zero.

2.4 Assumptions

Core bundle. The affinity and cutoff theorems formally take $A : \text{CoreEconomicAssumptions } P$. Within that bundle their proof terms access `r_pos`, `lambda_nonneg`, and `sigma_pos`. Economic admissibility additionally accesses `c_pos` and `beta_lt_one`, as well as equations (1)–(2) and equilibrium $\theta > 0$.

Matching bundle. Paper-facing admissibility and capstone theorems formally take

$$M : \text{MatchingAssumptions } P.$$

Their proof terms access only `vacancyMeetingRate_pos`. The monotonicity fields remain part of the bundled theorem premise even though they are not referenced by the proof term.

Shock bundle. Wrappers taking $D : \text{ShockAssumptions } P$ access only `D.isProbability` in the M2 affinity and zero arguments. The remaining distributional fields are carried by the paper-level bundle but are not accessed by those proof terms.

Thus the formal paper-facing premises include the complete bundles where shown in the signatures. Within them, the proofs do not access atomlessness, upper-support maximality, shock moment normalization, matching monotonicity, or any elasticity or differentiability restriction.

2.5 Equation (12) dependency

`equation12` follows from the affine-difference theorem evaluated at the derived cutoff. The separate zero theorem is required to simplify (12) to

$$S(\varepsilon) = \frac{\sigma}{r + \lambda}(\varepsilon - \varepsilon_d).$$

2.6 Interpretation, limitations, and next dependency

The positive slope means that higher idiosyncratic productivity raises joint match surplus by the constant factor $\sigma/(r + \lambda)$. The unique zero is the reservation productivity: firms destroy matches below it, are indifferent at it, and retain matches at or above it.

These results are conditional on an existing **ValueEquilibrium**. M2 does not prove existence of that equilibrium or of the joint pair (θ, ε_d) . It proves a unique zero on $\mathbb{R}$ and that the zero lies below ε_u , but not that it belongs to the topological support of the shock law. No lower-support assumption is present; the cutoff may lie below the lower effective support, making destruction probability zero.

Human-review conclusion. The surplus-affinity proof is non-circular, and the cutoff is derived from actual equilibrium values. Mathematical zero existence and economic admissibility remain separate; equations (1)–(2) are used only for admissibility. No M3 result is assumed or proved. The reviewed capstone explicitly identifies every surplus zero with **reservationCutoff**.

M3 may use the active-surplus positive-part formula to derive the continuation-value identity and the job-destruction and job-creation equations. No such integral identity is proved in M2.

Review space: